

Data Handoff — Dalibor → Ralph

Project 11_02 — Near Real-Time Financial Forecasting using Amazon SageMaker

Overview

Dalibor's job (steps 3–4) is to take the raw price and news messages from Ilyas's SQS queues, clean them, and produce a single **forecast-ready file in S3**.

Ralph (steps 5–6) then points Amazon SageMaker at that file to train and generate predictions.

Ralph needs **one clean file per ticker**, in a format Amazon SageMaker accepts directly — no extra cleaning on Ralph's side.

Context: What Ilyas gives Dalibor (recap)

Ilyas's Lambda pushes JSON messages into two SQS queues every X minutes:

proj1102-prices-queue — one message per price fetch:

```
{
  "ticker": "AAPL",
  "timestamp": "2026-04-18T10:00:00Z",
  "price": 185.42,
  "volume": 1234567,
  "source": "yahoo_finance"
}
```

proj1102-news-queue — one message per news headline:

```
{
  "ticker": "AAPL",
  "timestamp": "2026-04-18T10:00:00Z",
  "headline": "Apple announces record iPhone sales",
  "url": "...",
  "source": "newsapi"
}
```

Dalibor's Glue job consumes these, cleans them, and writes the result to S3.

What Dalibor gives Ralph

One clean Parquet file per ticker, stored in S3.

S3 path (agreed)

```
s3://proj1102-data/forecast-ready/AAPL.parquet
s3://proj1102-data/forecast-ready/TSLA.parquet
s3://proj1102-data/forecast-ready/BTC-USD.parquet
s3://proj1102-data/forecast-ready/ETH-USD.parquet
```

File format

- **Format:** Parquet (preferred) or CSV (acceptable)
- **One file per ticker**, overwritten or appended each Glue run
- **Compression:** Snappy (default for Parquet)

Schema (column names are **FIXED** — Amazon SageMaker is strict)

Column	Type	Required?	Example	Notes
item_id	string	yes	AAPL	Ticker name — must match across all rows of the file
timestamp	string ISO-8601	yes	2026-04-18T10:00:00Z	UTC, no missing intervals
target_value	float	yes	185.42	The closing price for that timestamp
sentiment_score	float (-1 to +1)	optional	0.87	Filled in once Ralph's Comprehend Lambda is running. Can be null/empty at first

Column names must be **exactly** `item_id`, `timestamp`, `target_value`, `sentiment_score`. Any rename will cause the SageMaker import to fail the import.

Example file content

s3://proj1102-data/forecast-ready/AAPL.parquet (shown as table):

item_id	timestamp	target_value	sentiment_score
AAPL	2026-04-18T10:00:00Z	185.42	0.87
AAPL	2026-04-18T10:05:00Z	185.51	0.65
AAPL	2026-04-18T10:10:00Z	185.38	null
AAPL	2026-04-18T10:15:00Z	185.62	0.42
AAPL	2026-04-18T10:20:00Z	185.70	0.50

- One row = one time step (every 5 minutes in this example)
- `sentiment_score` can be null if no news arrived for that interval
- No gaps in `timestamp` (no missing 5-minute windows) — fill with the previous price if no new fetch arrived

Cleaning rules Dalibor's Glue job must apply

1. **Deduplication:** if Ilyas's Lambda accidentally pushes the same (`ticker`, `timestamp`) twice, keep only one row.

2. **Time alignment:** round timestamps to the nearest 5-minute (or agreed) interval.
3. **Missing intervals:** if no price arrived for a slot, fill with the previous price (forward-fill).
4. **Sentiment join:** join with Ralph's sentiment CSV (in `s3://proj1102-data/sentiment/`) on (ticker, timestamp). If no sentiment for that timestamp, leave `sentiment_score` as null.
5. **Sort:** ascending by timestamp.
6. **No duplicates of (item_id, timestamp) allowed** — SageMaker will reject the import otherwise.

Amazon SageMaker import requirements (why these rules exist)

Amazon SageMaker is strict. The Parquet file Dalibor produces must satisfy:

- **Exact column names** as above (`item_id`, `timestamp`, `target_value`, `sentiment_score`)
- **Timestamp format** ISO-8601: `yyyy-MM-ddTHH:mm:ssZ` (no microseconds, no timezone offsets like `+02:00`)
- **No missing target_value** — every row must have a price
- **Numeric columns must be floats**, not strings
- **At least 300 historical data points** per `item_id` for training (so we need ~25 hours of 5-minute data per ticker before the first training run)

Sentiment data location (for reference)

Ralph's Lambda writes sentiment scores to:

`s3://proj1102-data/sentiment/AAPL/2026/04/18/sentiment_2026-04-18T10-00-00Z.csv`

CSV format:

ticker	timestamp	sentiment_score	confidence
AAPL	2026-04-18T10:00:00Z	0.87	0.92
AAPL	2026-04-18T10:05:00Z	0.65	0.81

Dalibor's Glue job reads from this folder to enrich the Parquet file.

Full data flow (summary)

Ilyas (Lambda)

--> SQS: proj1102-prices-queue (price JSON)

--> SQS: proj1102-news-queue (news JSON)

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Dalibor (Glue)

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|--> reads sentiment from `s3://proj1102-data/sentiment/`

|

+++> writes `s3://proj1102-data/forecast-ready/{ticker}.parquet`
 (`item_id`, `timestamp`, `target_value`, `sentiment_score`)

|

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Ralph (SageMaker)
-> imports the file, trains predictor, generates forecasts

What Dalibor can start building now (with dummy data)

Dalibor does not need to wait for Ilyas's real SQS messages. He can write a small script that generates fake price + sentiment rows and produces a sample Parquet file in the agreed format. This lets Ralph test the SageMaker import in parallel.

Dummy rows for testing:

```
import pandas as pd

df = pd.DataFrame([
    {"item_id": "AAPL", "timestamp": "2026-04-18T10:00:00Z", "target_value": 185.42, "sentiment_s"},
    {"item_id": "AAPL", "timestamp": "2026-04-18T10:05:00Z", "target_value": 185.51, "sentiment_s"},
    {"item_id": "AAPL", "timestamp": "2026-04-18T10:10:00Z", "target_value": 185.38, "sentiment_s"},
    {"item_id": "AAPL", "timestamp": "2026-04-18T10:15:00Z", "target_value": 185.62, "sentiment_s"}
])
df.to_parquet("AAPL.parquet")
```

Upload AAPL.parquet to s3://proj1102-data/forecast-ready/AAPL.parquet and Ralph can immediately try the SageMaker import on it.

Things to freeze between Dalibor and Ralph (agree before coding)

Item	Agreed value
S3 bucket	proj1102-data
Forecast-ready S3 path	s3://proj1102-data/forecast-ready/{ticker}.parquet
Sentiment S3 path	s3://proj1102-data/sentiment/{ticker}/...
File format	Parquet (Snappy compression)
Time granularity	5-minute bars
Required columns	item_id, timestamp, target_value, sentiment_score
Timestamp format	ISO-8601 UTC (e.g. 2026-04-18T10:00:00Z)
One file per ticker	Yes
Glue run frequency	Hourly (to be confirmed)

Any change to the above must be announced to the group immediately — a renamed column or path breaks both Dalibor's writer and Ralph's SageMaker import.

Quick checklist for Dalibor

- ☐ S3 bucket proj1102-data created in agreed region
- ☐ Glue job reads from both SQS queues (or from raw S3 dump)
- ☐ Output schema matches the table above exactly

- ☐ Parquet file written to `s3://proj1102-data/forecast-ready/{ticker}.parquet`
- ☐ Tested with dummy data first
- ☐ No duplicate (`item_id`, `timestamp`) rows
- ☐ No missing intervals (forward-fill applied)
- ☐ Sentiment column joined when available, null otherwise
- ☐ Glue job ARN shared with Arthur (for Step Functions)